

NGUYEN HOANG THANH PHAN

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EDUCATION

Wilfrid Laurier University, Waterloo, Ontario

Bachelor of Science in Computer Science, Co-op Program

Relevant Coursework: Internet Computing, Computer Networks, Software Engineering, Database Systems, Operating Systems, Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence

September 2024 – August 2028

Cumulative GPA: 12.0/12.0

PROJECTS

To-Do List Web Application — Java, Spring Boot, MySQL, Docker, Railway, Render, MS Azure

[GitHub](#) | [Demo](#) | April 2026

- Developed a full-stack task management web application using Spring Boot MVC, delivering 10+ RESTful APIs with JWT authentication for secure user access and JSON-based client-server communication
- Designed a normalized MySQL relational database schema across 3+ relational tables for users, tasks, and tags, implementing optimized SQL queries to support task filtering and retrieval
- Containerized the frontend, backend, and MySQL database using Docker Compose, reducing project setup from multiple manual configuration steps to a single-command deployment
- Created 10+ unit and integration tests with JUnit and Mockito, improving reliability and consistency of authentication and task-management workflows by validating core business logic and API behaviour
- Deployed a multi-cloud full-stack application with Netlify, Render, and a managed MySQL database on Microsoft Azure, configuring secure environment variables and cloud database connectivity for production

Dynamic Parking in Urban Networks — Python, NumPy, Reinforcement Learning

[GitHub](#) | April 2026

- Built a Q-learning-based parking allocation solution on weighted urban graph networks to optimize vehicle routing under dynamic traffic conditions
- Automated large-scale simulation experiments using Python, NumPy, and multiprocessing to execute parallel simulation runs, significantly improving data collection efficiency
- Analyzed experimental results by exporting simulation data to Excel and creating performance visualizations to evaluate learning behavior, convergence trends, and algorithm effectiveness

E-Commerce System — Python, Flask, MySQL

[GitHub](#) | [Demo](#) | December 2025

- Built an e-commerce web application backend with 10+ REST APIs supporting authentication, shopping cart, and checkout workflows for secure transaction processing
- Analyzed business requirements and led the implementation of a MySQL relational database across 4+ tables for customers, products, orders, and reviews
- Collaborated within a 12-week Agile team, contributing to sprint planning, feature implementation, code reviews, testing, and iterative software delivery

Supply Chain and Logistics Management Database — MySQL

[GitHub](#) | December 2025

- Designed a normalized relational database with 20+ tables to model warehouse, supplier, retailer, inventory, and shipment operations for end-to-end supply chain management
- Translated business requirements into complex SQL queries to analyze inventory levels, forecast stock demand, and identify order trends, supporting operational planning and reporting
- Produced ER diagrams and technical design documentation to improve database maintainability and facilitate collaboration among project contributors

SKILLS

Languages: Java, Python, C/C++, SQL, HTML5, CSS3, JavaScript

Frameworks & Libraries: Spring Boot, Spring Data JPA, Hibernate, Flask

Databases: MySQL, Relational Database Design

Testing: JUnit, Mockito, Spring Boot Test, REST API Testing, Unit Testing, Integration Testing

Microsoft Office: Word, Excel, PowerPoint, Outlook, VBA

Cloud & Deployment: Microsoft Azure, Render, Netlify

DevOps & Tools: Maven, Docker, Docker Compose, Git, GitHub, Postman

AWARDS & ACHIEVEMENTS

Dean's Honour Roll - Faculty of Science, Wilfrid Laurier University

2024 – 2025

HackCanada Participant

March 2026

Developed a skin disease classification web app using FastAPI, React, and a machine learning model in 72 hours.